iSAID Vehicle: Text SAM vs Detection-Guided SAM

- Goal: compare text-only, detector-guided, oracle-box, zero-shot, and remote-sensing fine-tuned SAM-sty

- Target: merged vehicle class from iSAID Small_Vehicle + Large_Vehicle.

- Eval design: 128 overhead urban tiles, 32 per overlap / mask-area stratum.

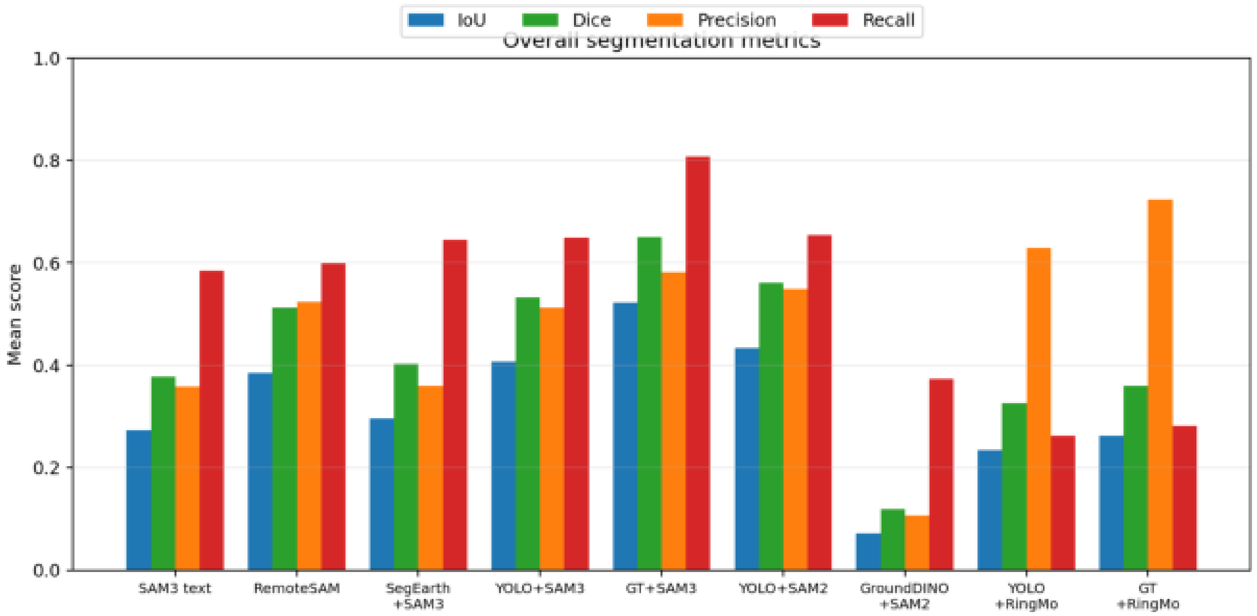
- Low/high area split uses eval median mask-area ratio 0.004528.

- Final inference was CPU-only; no GPU was used during this reporting run.

YOLO26x checkpoint	Value
Best epoch	51
Stopped epoch	74
Elapsed training	4h 55m 55s
mAP50 / mAP50-95	0.6097 / 0.3377
Precision / Recall	0.7356 / 0.5850

Overall Results

Mean scores over 128 eval images

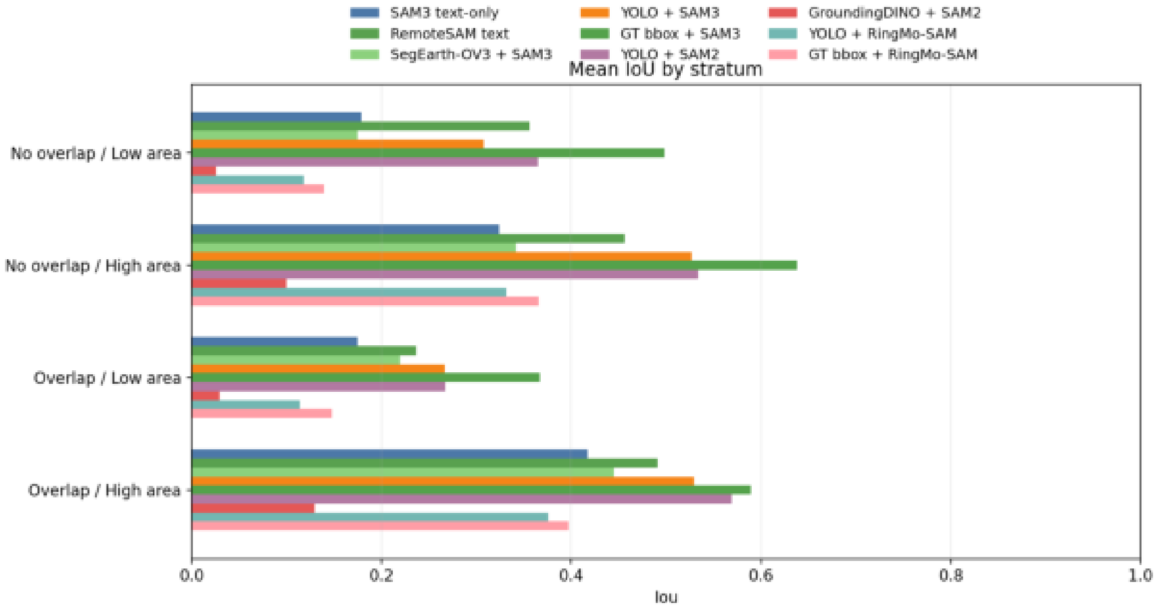


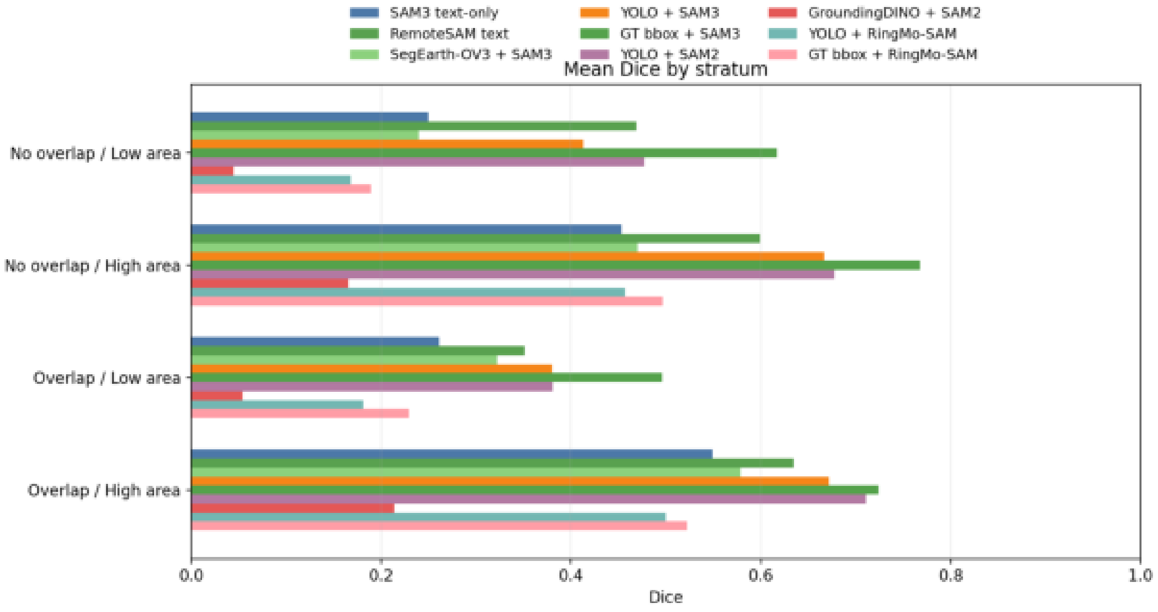
Pipeline	IoU	Dice	Prec.	Recall
SAM3 text	0.2739	0.3782	0.3589	0.5850
RemoteSAM	0.3850	0.5132	0.5244	0.5993
SegEarth+SAM3	0.2952	0.4027	0.3599	0.6449
YOLO+SAM3	0.4076	0.5328	0.5130	0.6485
GT+SAM3	0.5230	0.6510	0.5826	0.8072
YOLO+SAM2	0.4336	0.5615	0.5494	0.6549
GroundDINO+SAM2	0.0713	0.1196	0.1059	0.3734
YOLO+RingMo	0.2349	0.3266	0.6292	0.2630
GT+RingMo	0.2625	0.3592	0.7247	0.2815

- Detection-guided SAM improves over text-only in every stratum.
- GT boxes + SAM3 is the upper-bound detection case.
- YOLO + SAM2 is slightly ahead of YOLO + SAM3 overall in this run.
- RemoteSAM text is the strongest box-free text/referring baseline.
- SegEarth-OV3 tests SAM3 semantic/instance fusion without boxes.
- GroundingDINO + SAM2 is a zero-shot text-to-box baseline.
- RingMo-SAM is a remote-sensing fine-tuned box-constrained baseline.

Stratified View

2x2 split: bbox overlap yes/no x target mask area low/high





- Low-mask-area strata are the hardest: small vehicles punish over-segmentation and missed detections.

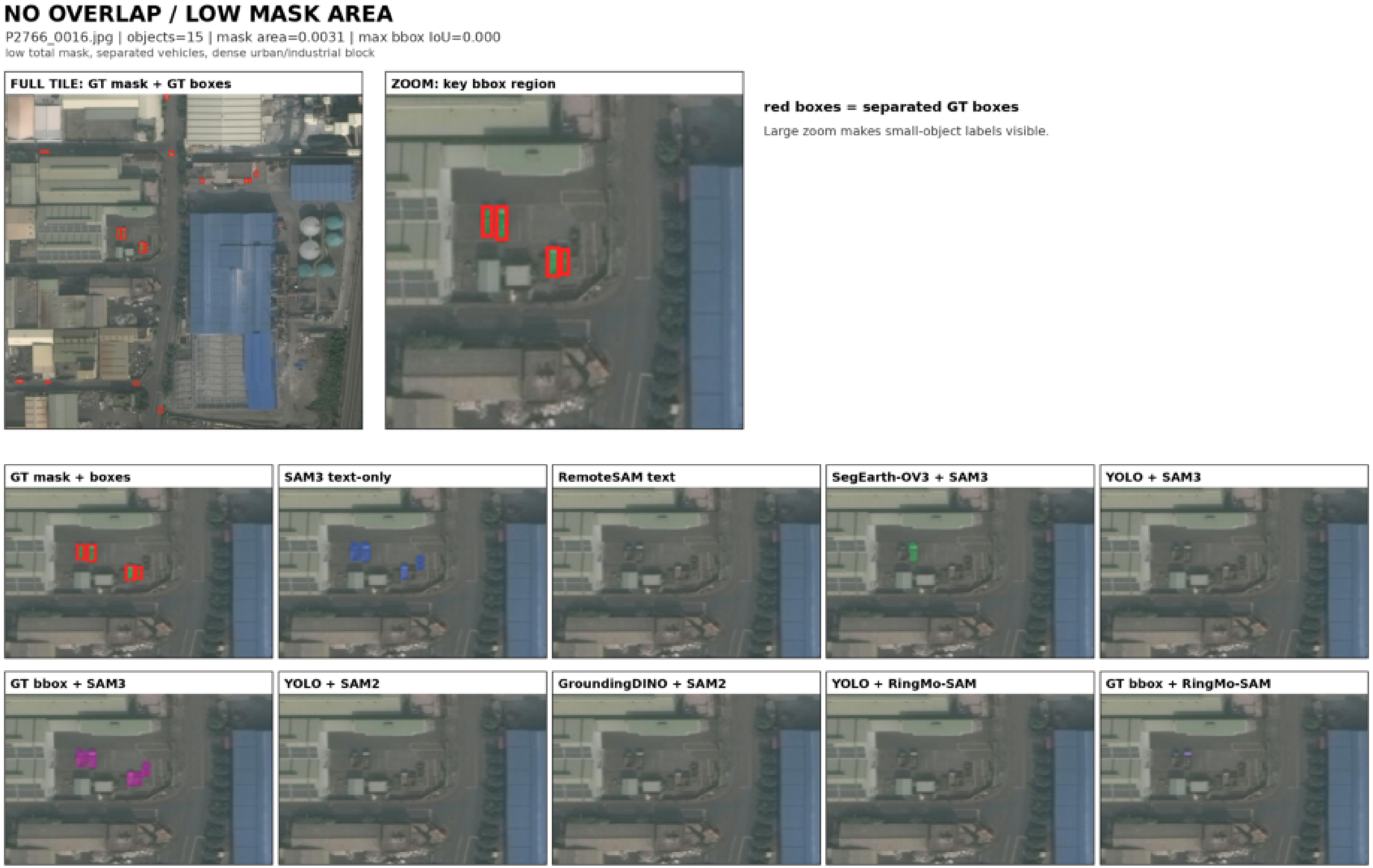
- High-mask-area strata give guided methods more signal and higher IoU/Dice.

Stratified Metrics Table

Stratum	Pipeline	IoU	Dice
No overlap / Low area	SAM3 text-only	0.1789	0.2495
No overlap / Low area	RemoteSAM text	0.3563	0.4687
No overlap / Low area	SegEarth-OV3 + SAM3	0.1746	0.2396
No overlap / Low area	YOLO + SAM3	0.3078	0.4128
No overlap / Low area	GT bbox + SAM3	0.4980	0.6172
No overlap / Low area	YOLO + SAM2	0.3647	0.4771
No overlap / Low area	GroundingDINO + SAM2	0.0262	0.0447
No overlap / Low area	YOLO + RingMo-SAM	0.1184	0.1681
No overlap / Low area	GT bbox + RingMo-SAM	0.1391	0.1894
No overlap / High area	SAM3 text-only	0.3244	0.4530
No overlap / High area	RemoteSAM text	0.4563	0.5989
No overlap / High area	SegEarth-OV3 + SAM3	0.3413	0.4702
No overlap / High area	YOLO + SAM3	0.5269	0.6669
No overlap / High area	GT bbox + SAM3	0.6380	0.7677
No overlap / High area	YOLO + SAM2	0.5339	0.6779
No overlap / High area	GroundingDINO + SAM2	0.0997	0.1660
No overlap / High area	YOLO + RingMo-SAM	0.3315	0.4570
No overlap / High area	GT bbox + RingMo-SAM	0.3656	0.4963
Overlap / Low area	SAM3 text-only	0.1748	0.2614
Overlap / Low area	RemoteSAM text	0.2365	0.3513
Overlap / Low area	SegEarth-OV3 + SAM3	0.2201	0.3225
Overlap / Low area	YOLO + SAM3	0.2663	0.3798
Overlap / Low area	GT bbox + SAM3	0.3668	0.4956
Overlap / Low area	YOLO + SAM2	0.2671	0.3804
Overlap / Low area	GroundingDINO + SAM2	0.0295	0.0540
Overlap / Low area	YOLO + RingMo-SAM	0.1139	0.1812
Overlap / Low area	GT bbox + RingMo-SAM	0.1478	0.2291
Overlap / High area	SAM3 text-only	0.4174	0.5489
Overlap / High area	RemoteSAM text	0.4911	0.6341
Overlap / High area	SegEarth-OV3 + SAM3	0.4447	0.5786
Overlap / High area	YOLO + SAM3	0.5293	0.6715
Overlap / High area	GT bbox + SAM3	0.5892	0.7235
Overlap / High area	YOLO + SAM2	0.5690	0.7106
Overlap / High area	GroundingDINO + SAM2	0.1297	0.2136
Overlap / High area	YOLO + RingMo-SAM	0.3760	0.5003
Overlap / High area	GT bbox + RingMo-SAM	0.3976	0.5221

no overlap / low mask area

Curated qualitative example. Card shows full context, zoomed GT boxes, and model masks.

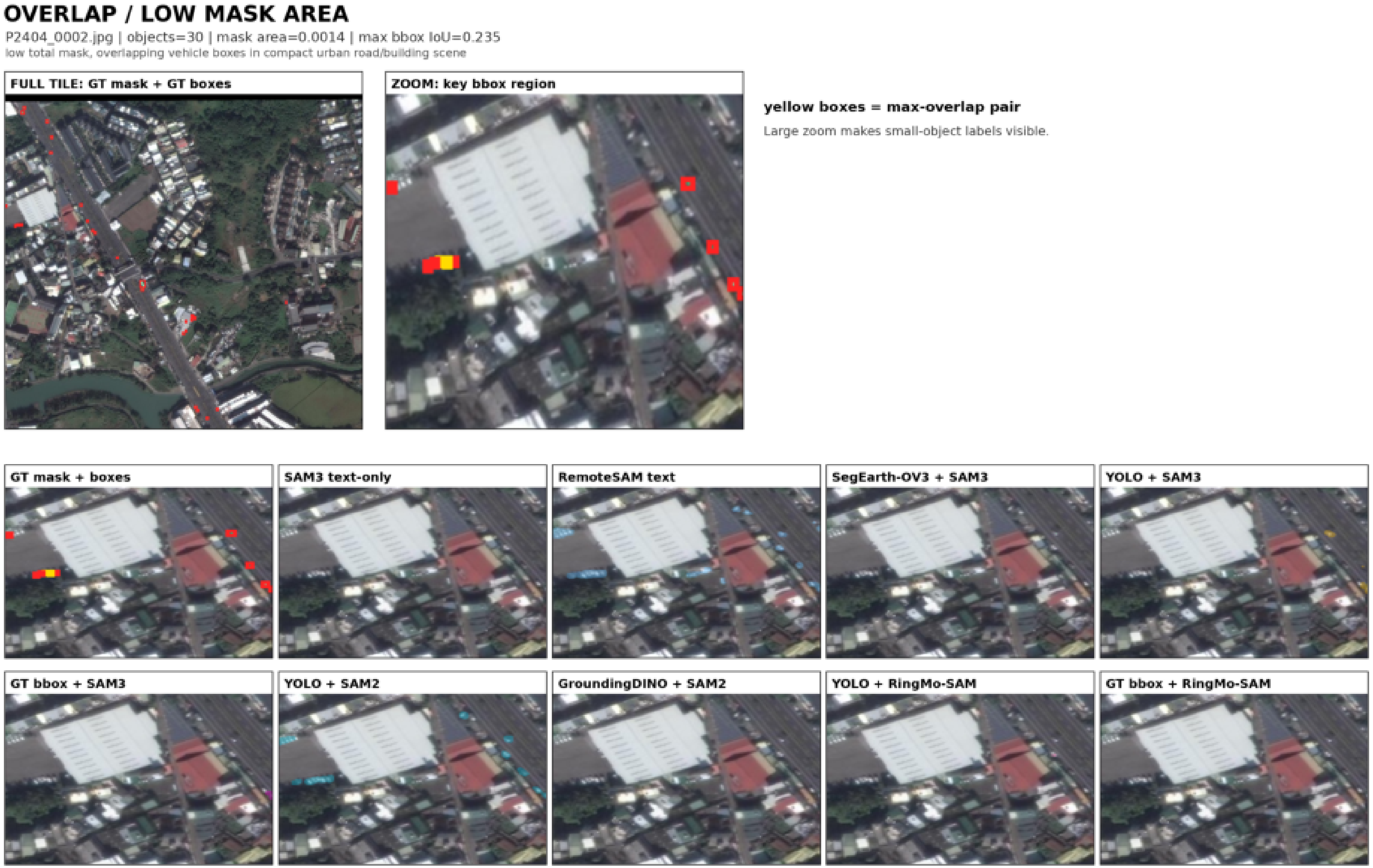


no overlap / high mask area

Curated qualitative example. Card shows full context, zoomed GT boxes, and model masks.

overlap / low mask area

Curated qualitative example. Card shows full context, zoomed GT boxes, and model masks.



overlap / high mask area

Curated qualitative example. Card shows full context, zoomed GT boxes, and model masks.

Takeaways

- YOLO guidance matters: YOLO + SAM3 beats SAM3 text-only in every 2x2 stratum.
- RemoteSAM text narrows the box-free gap but remains below YOLO + SAM2 overall.
- SegEarth-OV3 increases recall but tends to over-mask low-area scenes.
- Detection quality matters: GT bbox + SAM3 remains the best IoU/Dice case overall and by stratum.
- SAM2 is competitive with the same YOLO boxes and slightly better overall here.
- GroundingDINO + SAM2 tests text-to-box prompting separately from YOLO training.
- RingMo-SAM tests a remote-sensing fine-tuned decoder under fixed YOLO/GT boxes.
- Text-only SAM3 over-segments small-object scenes, especially low-mask-area tiles.
- The selected examples are balanced across overlap/no-overlap and low/high total vehicle mask area.